

Executive Summary

PASS RATE 28% target 90%	TOTAL TESTS 168 executed	PASSED 48 29% of total	FAILED 120 71% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 28% | Target: 90% | Gap: +62%

The Revenue Management System Kenya backend API testing reveals significant stability and reliability challenges, with an overall pass rate of 28% against a 90% target, indicating substantial issues in endpoint functionality and response behavior. The API Automation suite achieved 27.5% pass rate (22/80 tests) while Backend Smoke Validation reached 29.5% (26/88 tests), suggesting systemic issues across both critical and comprehensive test scenarios. The consistently low performance across both test suites indicates fundamental backend service issues rather than isolated endpoint failures, requiring immediate technical investigation and remediation to ensure the revenue management system can reliably process transactions and maintain data integrity for the Kenya market operations.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	76	22	54	29%	FAIL
2	BE Smoke Validation	88	26	62	30%	FAIL
TOTAL		168	48	120	28%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#49	05 Jul	168	28%	120	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	48	28.6%	180ms	Within SLA
Requests Failed	120	71.4%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

Severity	Count	%
Critical	6	5%
High	8	7%
Medium	4	3%
Low	2	2%

QUALITY INDICATORS

Indicator	Value	Target
Endpoint Coverage	88%	85%
Execution Rate	100%	100%
Contract Validation	95%	90%
Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate root cause analysis on the top 10 most frequently failing endpoints to identify common failure patterns	Immediate	P1
2	Review backend service logs and monitoring data to identify performance bottlenecks, timeout issues, or resource constraints	3 days	P1
3	Verify API response schemas and data contracts align with expected formats across all failing endpoints	5 days	P2
4	Implement enhanced error handling and retry mechanisms for critical revenue processing endpoints	1 week	P2
5	Deploy additional backend monitoring and alerting for real-time visibility into service health metrics	1 week	P2
6	Prioritize fixes for smoke test failures as these represent core functionality required for basic system operations	2 weeks	P3

KEY FINDINGS

1	Backend services demonstrate 72% failure rate across 168 automated test scenarios, indicating widespread API instability
2	API Automation suite shows 58 endpoint failures suggesting core service functionality issues or incorrect response formatting
3	Smoke validation failures (62 of 88) indicate critical path operations are not functioning as expected
4	Consistent failure patterns across both test suites point to systemic backend architecture or configuration problems
5	Current 28% pass rate represents a 62 percentage point gap from the 90% reliability target

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation



29%

